

Team Project

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$ echo "Data Sciences Institute"
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Today

1. Review from last week.
2. Communicating business impact.

Review

Review

- Last week, we focused on several key topics:
 - i. How to present this project in your portfolio.
 - ii. How to effectively develop a business case for your project.
 - iii. How to work together as a team.
 - iv. How to hold effective standup meetings.

Project Plan

- You have each received feedback on your Project Plan in the form of a GitHub issue. You should incorporate this feedback into your README this week.
- When submitting your final project, ensure that you have a section in the README for changes you have made to your original project plan.

Keep in Mind

1. **Business Relevance** → Your project should address a real problem or opportunity in your chosen industry, and your results should provide concrete recommendations for your stakeholders.
2. **Good Code & Structure** → Code should be well-commented, clean, and follow a logical structure. It should be easy to read and maintain.
3. **Strong Documentation & Presentation** → The README should be clear, well-written, and explain the dataset, findings, and methodology. The project should be easy to understand for both technical and non-technical reviewers.

Keep in Mind

- 4. **Application of Module Teachings** → Projects should showcase key technical skills, such as regression modeling, deep learning models, data visualizations, or strong analysis of sampling techniques.
- 5. **Effective Team Collaboration** → Teams should follow best practices for Git (small commits, branches, pull requests) and actively participate in stand-ups and progress updates.

Crafting a Comprehensive Main README File

- **Purpose & Overview:** Introduce the project with essential details, concise description and a project objective.
- **Goals & Objectives:** Articulate what the project aims to achieve. Include any changes made to the original project plan here.
- **Techniques & Technologies:** Highlight the tools and methods used.
- **Key Findings & Instructions:** Summarize outcomes and provide setup instructions.
- **Visuals & Credits:** Enhance with visuals; acknowledge contributors.

This Week's Schedule

Day 6 (Tues): Content delivery, co-work + standups.

Day 7 (Wed): Co-work + standups.

Day 8 (Thurs): Co-work + standups.

Day 9 (Fri): Co-work.

Day 10 (Sat): Project showcase.

Standups

- Each day, a member of the DSI instructional team will guide your team through a standup.
- Stand-ups are quick, structured check-ins that help teams stay on track and remove obstacles.
- The goal is not to compete for who did the most work, it's to ensure the entire team is working effectively and efficiently.
- This is a great opportunity to help your teammates and resolve blockers early.
- Standups should take no more than 10 minutes.

Communicating & Presenting Your Findings

Communicating & Presenting Your Findings

- Even if you have conducted a thorough analysis and found extremely useful results, your project will only be considered impactful if you communicate your findings effectively.

Focus on:

1. Contextualizing your project.

- Clearly describe the business context for your project. What is the gap that you are filling with this work? Where does it fit in the wider industry landscape?

2. Quantifying the business impact.

- What are the key takeaways for your stakeholders? Focus on actionable items.

3. Making complex concepts accessible for your audience.

- How can you communicate technical concepts in a way that everyone can understand?
- Don't explain all your methodological details right away. Start with the big concepts and if the audience is interested in details they will ask.

Project Context

- Try to tell a story!
- Start with the big picture, then slowly focus in on your specific project context.

Business Impact

- Quantify everything that you can!
- Recommend actions that your audience can take to generate business value.

Business Impact

Showcase your project's business value by applying specialized skills:

-  **Data Science Stream:** Apply the techniques learned in your sampling and visualization modules to deliver impactful data insights.
-  **Machine Learning Stream:** Demonstrate model efficiency and algorithm optimization with detailed performance graphs and clear pipeline visualizations.

Ensure your presentations translate complex data into clear, actionable business insights.



Examples of Quantifying Impact

Retail Example

*"Showed that costs could be reduced by optimizing inventory management. Data was cleaned and aggregated using SQL queries to identify sales patterns. Visualizations in Tableau helped to **forecast demand, leading to cost savings of 5%.**"*

Examples of Quantifying Impact

Marketing Example

Showed that customer engagement could be increased by 8% with a new segmentation scheme. Feature engineering was conducted to create meaningful customer segments based on purchase history and demographics.

Making Complex Concepts Accessible

Communicating complex data science concepts clearly is essential for audiences without technical backgrounds, including recruiters and HR teams.

Strategies to Enhance Understandability:

- **Use Analogies and Metaphors:** Bridge understanding gaps with relatable comparisons.
- **Incremental Explanation:** Simplify concepts step-by-step.
- **Visual Aids:** Employ diagrams and infographics for clarity.

Clear communication ensures your projects are comprehensible to all, enhancing your professional appeal and broadening project impact.



Example of Making Complex Concepts Accessible

Before:

"Improved patient outcomes by developing a predictive model using Python and scikit-learn. Data was preprocessed using pandas to handle missing values and standardize features. Logistic regression was selected through cross-validation due to its interpretability and performance."

Example of Making Complex Concepts Accessible

After:

“Improved patient outcomes by developing a predictive model using Python and scikit-learn. Think of the predictive model as a way to forecast patient health based on past data, much like predicting the weather. Data was preprocessed using pandas to handle missing values and standardize features. This preprocessing is like organizing and cleaning up messy medical records to ensure all information is complete and consistent. Logistic regression was selected through cross-validation due to its interpretability and performance. Imagine trying different medical treatments and choosing the one that consistently gives the best results; that’s what cross-validation does for model selection.”

This approach effectively communicates complex concepts to both technical and non-technical audiences, ensuring clarity and comprehension across diverse stakeholders.

Project Evaluation & Showcase

Final Project Submission

- The DSI team will begin evaluating your projects and providing feedback in the form of a *Git Issue* next Monday, March 9th (this will likely take 1 week).
- We encourage you to record your personal videos after the Project Showcase, as you will likely get some good ideas on how to describe your project!

Final Project Submission

Your final project will be evaluated on the following criteria:

1. Each team member must have created a pull request, and reviewed and merged a different pull request.
2. (a) For **Data Science** teams, your project must include visualizations that presents new insights into the chosen dataset.

(b) For **Machine Learning** teams, your project must include at least one machine learning model that you have developed and implemented to obtain new insights.

Final Project Submission

3. In addition to the project proposal from Week 1, each project's README should describe the final outcome of the project, the key business takeaways, and describe your team's approach to working collaboratively. It should also demonstrate thoughtful consideration of the guiding questions listed in this repository's README.
 - *Use READMEs in subfolders to keep your main README concise and streamlined.*

Final Project Submission

4. Your project should be ***completely reproducible***. This means that anyone (e.g. the instructional team for marking, or potential employers for evaluating your work) should be able to completely recreate your analysis and results based on the instructions in your repository. This includes any software and data dependencies. You should test this by trying to regenerate your results from scratch.

Final Project Submission

5. Each team member must record a 3-5 minute video reflecting on your experience. You may each choose where to host your own video, however it should be public and a link to each team member's video should be included in your project README. This video is meant to be an asset to your portfolio, and should be available for prospective employers. Your videos should answer the following questions:

- What did you learn?
- What challenges did you face?
- How did you overcome those challenges?
- If you had more time, what would you add?
- What strengths do you bring to a team environment?

Project Submission

- Remember to clean up your repository.
 - Organize folders and include supporting documentation appropriately.
 - E.g. link from your main README to subfolders for additional information where applicable.
 - Remove unused files and folders.
 - Make sure your code is commented.

Project Showcase

- **The showcase will start at 09:00am ET sharp on Saturday, March 7th.**
- Each team will have 5 minutes to present your project. We will not be taking questions after each presentation, but if you do have questions for another team please write them down and we will go over them if there is time.
- It is up to your time how you wish to divide the presentation duties among team members.
- *Let us know by tomorrow (Wednesday) if none of your team members are available to attend, or if you have specific scheduling constraints.*

Project Showcase

- This day is for showcasing your progress on the project goals so far. It is about presenting what your project is about, what you've done, and what you plan to do next. It's also a chance to get feedback and ideas from your fellow participants.
- Put your best foot forward, as you may never get such a diverse audience of this size again. The audience is comprised of your fellow participants, who now have a good background in data science and machine learning, and have extensive expertise in their respective industries. They will provide valuable feedback.

Project Showcase

- This is not a lot of time, so you should not try to describe every step of your project. Instead, think of it as an "elevator pitch".
 - Assume that the audience is not an expert in your industry. Provide the required context.
 - Who are your intended stakeholders and why should they care about your project?
 - Explain your dataset in a way that your audience can understand.
 - BRIEFLY describe the tools/technologies that you used.
 - What are the key outcomes and takeaways? What is the business impact of your findings? Highlight one or two key visualizations or metrics.
 - If you had more time, what would your team have explored next?

Next Steps Post-Certification

As you transition from the certificate, continue to build and showcase your skills.

Strategies for Continuous Growth:

- **Active Portfolio:** Aim for several substantial projects to demonstrate a broad range of skills. Focus on quality over quantity.
- **Explore and Learn:** Continue exploring new areas within data science and machine learning to keep your knowledge current and dynamic.
- **Peer Collaboration:** Participate in group projects with fellow participants to enhance your skills and widen your professional network.

Actively applying for jobs should proceed in tandem with these activities.



Questions?